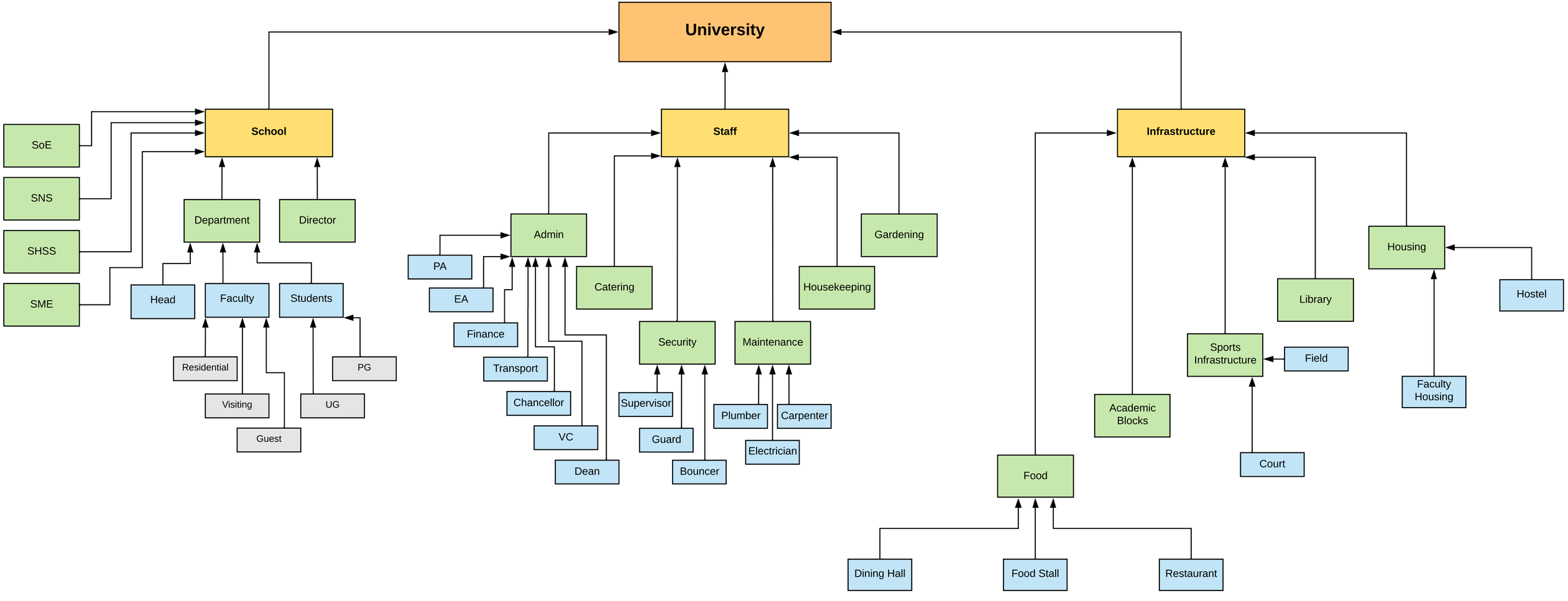
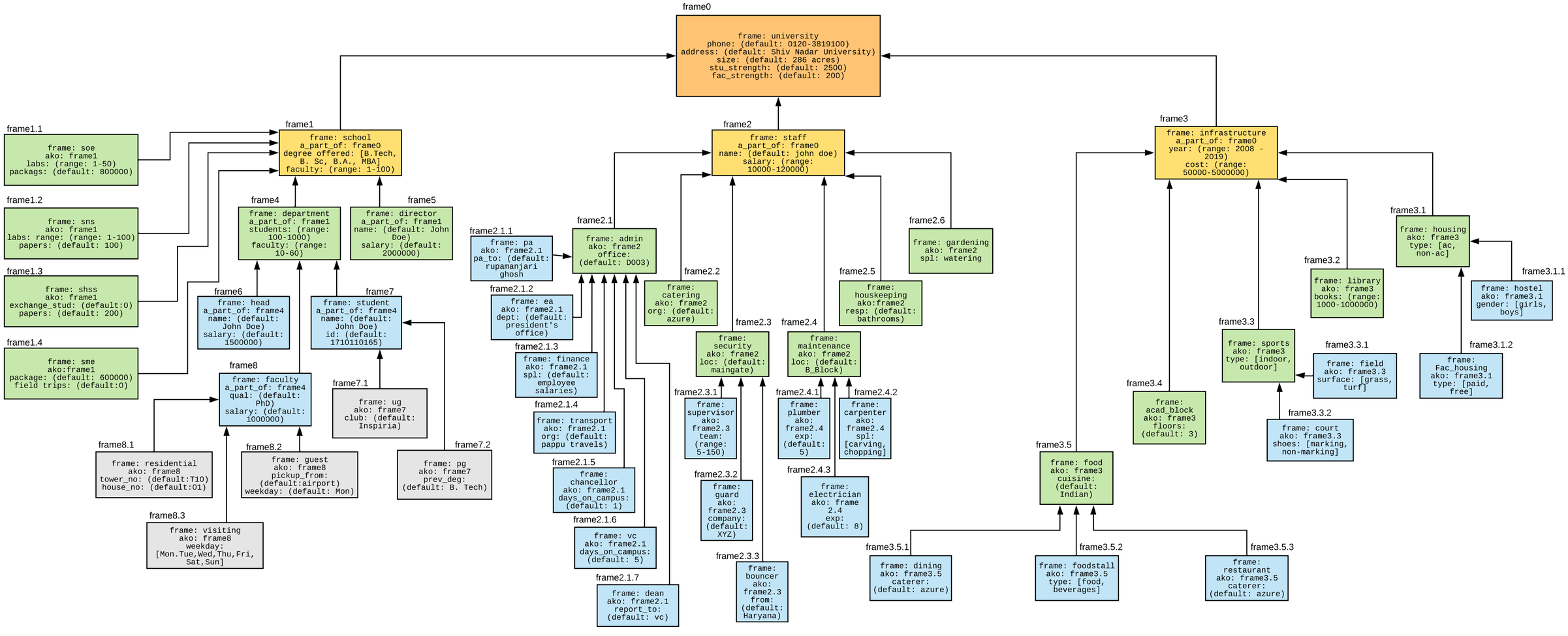
Artificial Intelligence (CSD311)

Assignment # 3

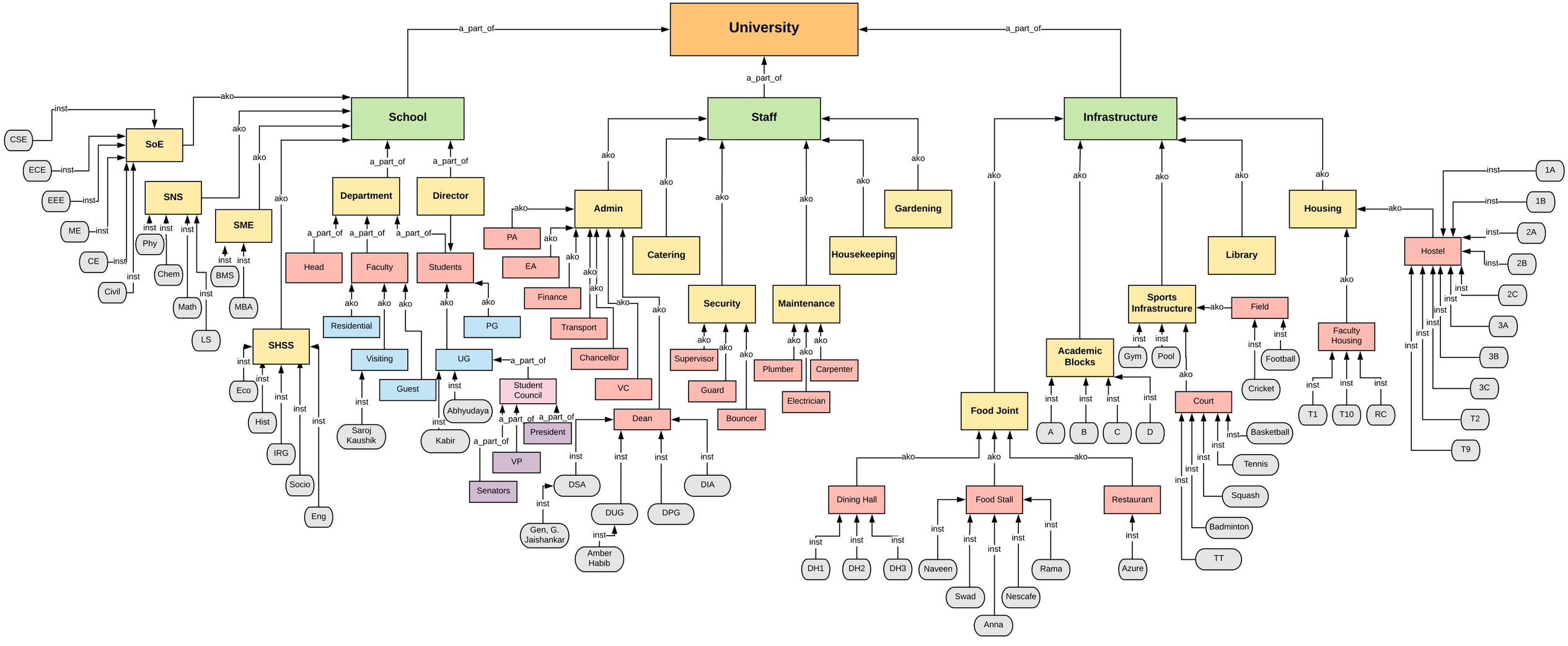
Frame Based System (FBS) for a University



Frames with slots for a University



Network of Frames (NOF) for a University



Q4. Module for insertion:

```
/*INSERT*/

/* IF FRAME ALREADY EXISTS*/

    insert_a_frame(F, _, _) :-

    frame(F, _),

    !,

    write("Frame already exists, CANNOT INSERT").


/* IF PARENT EXISTS*/

    insert_a_frame(F, P, List) :-

    frame(P,_),

    assertz(frame(F,List)),

    write("ADDED SUCCESSFULLY").


/* IF PARENT DOES NOT EXIST*/

    insert_a_frame(_, _,_) :-

    write("Parent not found, CANNOT INSERT").
```

Q5. Module for deletion:

```
/*DELETE*/
    delete_a_frame(F) :-
    frame(F,_),
    frame(Y, [(a_part_of, F) | _]),
    delete_a_frame(Y).

    delete_a_frame(F) :-
    frame(F, _),
    frame(Y, [(ako, F) | _]),
    delete_a_frame(Y).

    delete_a_frame(F) :-
    frame(F, Z),
    write("SUCCESSFULLY DELETED"),
    retract(frame(F, Z)).
```

Q6. QUERIES

Query 1:

display_all(X,Y).

Query 2:

show_all_info(kabir,X).

Query 3:

find_a_frame(soe,ako,X).

Query 4:

find_a_frame(kabir,id,X).

Query 5:

insert_a_frame(kka,ug,[(inst,(ug)),(name,("KABIR"))]).

Query 6:

delete_a_frame(kka).

Query 7:

delete_a_frame(does-not-exist).

Query 8:

find_a_frame(restaurant,caterer,X).

Query 9:

show_all_info(hostel,X).

Query 10:

insert_a_frame(kka,temp,[(inst,(ug)),name("KABIR")]).

CODE

```
delete_a_frame(A) :-
    frame(A, _),
    frame(B, [(a_part_of, A)|_]),
    delete_a_frame(B).
delete_a_frame(A) :-
    frame(A, _),
    frame(B, [(ako, A)|_]),
    delete_a_frame(B).
delete_a_frame(A) :-
    frame(A, B),
    write("SUCCESSFULLY DELETED"),
    retract(frame(A, B)).

save_file(A) :-
    open(A, write, B),
    with_output_to(B, listing),
    close(B).

search([(A, B)|_], A, B) :-
    !.
```

```
search([_|A], B, C) :-
    search(A, B, C).


show_all_info(A, B) :-
    frame(A, B).


frame(university, [nil,  (phone, default, "0120 3819100"),  (address, default, "Shiv Nadar
University"),  (size, default, "286 acres"),  (stu_strength, default, 2500),  (fac_strength,
default, 200)]).

frame(school, [(part_of, university),  (degree_offered, ["B.Tech", "B.Sc", "B.A", "MBA"]),
(teachers, range, 1, 100)]).

frame(staff, [(part_of, university),  (name, default, "John Doe"),  (office, default, "D003")]).

frame(infrastructure, [(part_of, university),  (year, range, 2008, 2019),  (cost, range, 50000,
5000000)]).

frame(soe, [(ako, school),  (labs, range, 1, 50),  (package, default, 800000)]).

frame(sns, [(ako, school),  (labs, range, 1, 100),  (papers, default, 100)]).

frame(shss, [(ako, school),  (exchange_stud, default, 0),  (papers, default, 200)]).

frame(sme, [(ako, school),  (field_trips, default, 0),  (package, default, 600000)]).

frame(department, [(part_of, school),  (students, range, 100, 1000),  (faculty, range, 10, 80)]).

frame(director, [(part_of, school),  (name, default, "John Doe"),  (salary, default, 2000000)]).

frame(head, [(part_of, department),  (name, default, "John Doe"),  (salary, default, 1500000)]).

frame(faculty, [(part_of, department),  (qual, default, "PhD"),  (salary, default, 1000000)]).

frame(student, [(part_of, department),  (name, default, "John Doe"),  (id, default, 1710110165)]).

frame(residential, [(ako, faculty),  (tower_no, default, "T10"),  (house_no, default, "001")]).
```

```
frame(guest, [(ako, faculty),  (pickup_from, default, "Airport"),  (weekday, default, "Mon")]).
frame(visiting, [(ako, faculty),  (weekday, ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"])]).
frame(ug, [(ako, student),  (club, default, "Inspiria")]).
frame(pg, [(ako, student),  (prev_deg, default, "B.Tech")]).
frame(admin, [(ako, staff),  (office, default, "D003")]).
frame(cetering, [(ako, staff),  (org, default, "Azure")]).
frame(security, [(ako, staff),  (loc, default, "Main Gate")]).
frame(maintenance, [(ako, staff),  (loc, default, "B Block")]).
frame(housekeeping, [(ako, staff),  (resp, default, "Bathrooms")]).
frame(gardening, [(ako, staff),  (spl, default, "Watering")]).
frame(pa, [(ako, admin),  (pa_to, default, "Rupamanjari Ghosh")]).
frame(ea, [(ako, admin),  (dept, default, "President's Office")]).
frame(finance, [(ako, admin),  (spl, default, "Employee Salaries")]).
frame(transport, [(ako, admin),  (org, default, "Pappu Travels")]).
frame(chancellor, [(ako, admin),  (days_on_campus, default, 1)]).
frame(vc, [(ako, admin),  (days_on_campus, default, 5)]).
frame(dean, [(ako, admin),  (report_to, default, "Vice Chancellor")]).
frame(supervisor, [(ako, security),  (team, range, 5, 150)]).
frame(guard, [(ako, security),  (company, default, "XYZ")]).
frame(bouncer, [(ako, security),  (from, default, "Haryana")]).
frame(plumber, [(ako, maintenance),  (exp, default, 5)]).
frame(electrician, [(ako, maintenance),  (exp, default, 8)]).
frame(carpenter, [(ako, maintenance),  (spl, ["Carving", "Chopping", "Repair"])]).
```

```
frame(housing, [(ako, infrastructure),  (type, ["AC", "Non-AC"])]).

frame(library, [(ako, infrastructure),  (books, range, 1000, 1000000)]).

frame(sports, [(ako, infrastructure),  (type, ["Indoor", "Outdoor"])]).

frame(acad_block, [(ako, infrastructure),  (floors, default, 3)]).

frame(food, [(ako, infrastructure),  (cuisine, default, "Indian")]).

frame(hostel, [(ako, housing),  (gender, ["Girls", "Boys", "Co-Ed"])]).

frame(fac_housing, [(ako, housing),  (type, ["Paid", "Free"])]).

frame(field, [(ako, sports),  (surface, ["Grass", "Turf"])]).

frame(court, [(ako, sports),  (shoes, ["Marking", "Non-Marking"])]).

frame(dining, [(ako, food),  (caterer, default, "Azure")]).

frame(foodstall, [(ako, food),  (type, ["Food", "Beverages"])]).

frame(restaurant, [(ako, food),  (caterer, default, "Azure")]).

frame(kabir, [(inst, ug),  (club, value, "Inspiria"),  (name, value, "Kabir Kanha Arora"),  (id,
value, 1710110165)]).

frame(kabir, [inst(ug), club(value, "Inspiria"), name(value, "Kabir Kanha Arora"), id(value,
1710110145)]).

frame(kabir, [inst(ug), club(value, "Inspiria"), name(value, "Kabir Kanha Arora"), id(value,
1710110145)]).

frame(kka, [(inst, ug),  (name, "KABIR")]).


display_all(A, B) :-

    frame(A, B).


find_a_frame(A, C, D) :-
```

```
    frame(A, B),
    search(B, C, D).
find_a_frame(A, C, D) :-
    frame(A, [(ako, B)|_]),
    find_a_frame(B, C, D).
find_a_frame(A, C, D) :-
    frame(A, [(a_part_of, B)|_]),
    find_a_frame(B, C, D).

insert_a_frame(A, _, _) :-
    frame(A, _),
    !,
    write("Frame already exists, CANNOT INSERT").
insert_a_frame(B, A, C) :-
    frame(A, _),
    assertz(frame(B, C)),
    write("ADDED SUCCESSFULLY").
insert_a_frame(_, _, _) :-
    write("Parent not found, CANNOT INSERT").
```